R13

Code No: 117FN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2023
MOBILE COMPUTING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What are the characteristics of communication devices?	[2]
b)	Explain about the MAC layer in DECT architecture.	[3]
c)	What are the drawbacks of FDMA?	[2]
d)	What is the basic purpose of DHCP?	[3]
e)	What is meant by timeout freezing in mobile transport layer?	[2]
f)	What are the advantages of Indirect TCP?	[3]
g)	Explain the functions of Pull based mechanisms in data delivery mechanisms.	[2]
h)	How do resolve synchronization issues?	[3]
i)	What is the function of transport layer in WAP?	[2]
j)	What are the features of Symbian OS?	[3]

PART – B

(50 Marks)

2. Describe the system architecture of GSM with neat sketch. [10]

OR

3. Explain about the protocol stack of GPRS. [10]

4.a) Explain about the IP – in – IP encapsulation.
b) Explain about Demand Assigned Multiple Access Protocol. [5+5]

OR

5.a) Explain about the packet delivery mechanism in mobile network layer.
b) Explain the frame format of an IEEE 802.11 PHY Frame using Frequency Hopping Spread Spectrum. [5+5]

6.a) Explain about the Mobile TCP.
b) Explain about the client server computing. [5+5]

OR

7.a) Explain about the caching invalidation mechanism.
b) What are the advantages of Snooping TCP? [7+3]

8.a) Explain the classification of data delivery mechanism.
b) Explain about the Index based method in selective tuning technique. [5+5]

OR

9. Briefly discuss about various types of data synchronization in mobile computing systems. [10]

10. Explain how to find the route from source to destination by using AODV protocol and also illustrate with an example. [10]

OR

11. Describe the architecture of Windows CE with neat sketch. [10]

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